



SPARE PARTS MANAGEMENT SYSTEM - PROJECT PHASES



COMPLETE PHASE BREAKDOWN

Based on our previous discussions and the project knowledge, here's the complete roadmap:

Phase 1: Core Infrastructure



Objective: Foundation & Authentication System

Features:

- **PSR-4 Autoloader** - Case-insensitive fallback for class loading
- **MVC Router** - Trailing slash tolerance, {id} parameter support
- **Authentication System** - Session management, login/logout
- **Bilingual Support** - English/Arabic with full RTL support
- **CSRF Protection** - All forms secured
- **Database Layer** - PDO abstraction with prepared statements
- **Responsive UI** - Modern gradient design, mobile-friendly
- **Security Features** - Password hashing, XSS protection, input



validation

Core Components:

```
/app/core/      # Framework classes
/app/config/    # Configuration management
/app/lang/      # Language files (en.php, ar.php)
/public/        # Web root with assets
/sql/           # Database schema and seeds
```

Phase 2: Masters CRUD



Objective: Master Data Management

Features:

- **Clients Management** - Company/Individual types with full contact details
- **Suppliers Management** - Complete supplier database with CRUD
- **Warehouses Management** - Multiple locations with responsible contacts
- **Products Management** - Detailed catalog with auto-generated codes
- **Dropdown Management** - Classifications, colors, brands, car makes/



models

Enhanced Features:

- **Auto Product Codes** - Generated based on classification (ENG0001, BDY0002)

- **Dependent Dropdowns** - Car models filter by car make (AJAX-powered)
- **Search & Pagination** - Find records across all modules
- **Warehouse Locations** - Track products across multiple warehouses
- **Client Profiles** - Tabs for quotes, orders, invoices, payments, balances
- **Stock Management** - Track quantities with low stock alerts

UI/UX Improvements:

- **Dropdown Navigation** - Masters menu with all modules
- **Enhanced Tables** - Sortable, searchable, paginated
- **Dynamic Forms** - Add/remove warehouse locations
- **Status Indicators** - Color-coded stock levels and statuses
- **Tab System** - Organized client/supplier details

Phase 3: Sales Flow

Objective: Complete Sales Workflow

Features:

- **Quotes Module** - Create, edit, approve/reject quotes
- **Sales Orders Module** - Convert quotes to orders, track
- shipping **Invoices Module** - Generate invoices from orders
- **Payments Module** - Record payments against invoices

Business Logic:



Advanced Calculations:

- **Line-level Tax/Discount** - Per item calculations
- **Global Tax/Discount** - Applied to entire quote/order
- **Multiple Tax Types** - Percentage or fixed amounts
- **Real-time Totals** - Automatic calculation updates

Stock Management:

- **Stock Reservations** - Reserved for quotes and orders

- **Stock Movements** - Complete audit trail
 - **Availability Checking** - Prevent overselling
 - **Multi-warehouse Support** - Track stock across locations
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Phase 4: Payments & Stock

Objective: Advanced Financial & Inventory Management

Features:

-  **Payment Methods** - Cash, bank transfer, check, credit card
 -  **Balance Tracking** - Client account balances
 -  **Stock Movements** - In/out tracking with references
 -  **Status Management** - Quote/order/invoice status workflows
 - **Purchase Orders** - Create POs for suppliers
 - **Goods Receipt** - Receive stock from suppliers
 - **Stock Adjustments** - Manual stock corrections
 - **Payment Terms** - 30/60/90 day terms
 - **Credit Limits** - Client credit management
 - **Aging Reports** - Outstanding receivables
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Phase 5: Email & PDF

Objective: Document Generation & Communication

Email Features:

- **SMTP Integration** - Send quotes/invoices via email
- **Email Templates** - Bilingual email templates
- **Attachment Support** - PDF attachments
- **Email Logging** - Track sent emails

PDF Generation:

- **FPDF Integration** - Generate PDF documents
- **Bilingual Templates** - English/Arabic PDF layouts
- **Company Branding** - Logo and styling
- **Document Types** - Quotes, orders, shipping order, invoices, statements

Templates:

- Quote PDF with Arabic/English layout

- Invoice PDF with payment terms
 - Statement PDF with aging details
 - Delivery note PDF for shipping
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Phase 6: Reports & Advanced Features

Objective: Business Intelligence & Advanced Operations

Reporting System:

- **Sales Reports** - By client, product, period
- **Inventory Reports** - Stock levels, movements, valuations
- **Financial Reports** - P&L, receivables aging, payment analysis
- **CSV Export** - All reports exportable to CSV

Advanced Features:

- **Client Portal** - Customers can view their quotes/invoices
- **Barcode Integration** - Product barcode scanning
- **Mobile App** - React Native mobile application
- **API Integration** - REST API for third-party integrations
- **Advanced Search** - Global search across all modules

Dashboard Enhancements:

- **Real-time Metrics** - Live sales/inventory dashboards
 - **Chart Integration** - Sales trends, top products
 - **KPI Tracking** - Monthly/quarterly performance
 - **Alert System** - Low stock, overdue payments
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System Strengths:

- **Scalable Architecture** - Modular design for easy expansion
- **Security-First** - CSRF protection, input validation, prepared statements
- **Business Logic** - Proper workflow enforcement and validation
- **International Ready** - Full Arabic support with RTL layout
- **Modern Tech Stack** - PHP 8+, MySQL 8, responsive CSS